Do LLMs Produce Regret-Like Language Under Deprivation Framing? A Controlled Behavioral Study Across Two Model Families

Jaeyeong CHOI
DGIST EECS
jaeyeong2022@dgist.ac.kr

***Abstract*—We investigate whether deprivation-framed prompts systematically elicit regret-associated linguistic patterns in large language models (LLMs). Using a $3 \times 3 \times 2$ controlled pilot design ($N = 216$; GPT-4o and Gemini-2.5-Flash), we find that deprivation framing produces substantially higher counterfactual-expression and regret-word rates compared to neutral framing. Exploratory Welch t-tests yield large effects (CF rate: $d = 0.78$, regret-word rate: $d = 0.76$), and a mixed-effects model (scenario as random intercept) confirms significant effects on semantic regret bias for both deprivation ($\hat{\beta} = 0.557$, $z = 5.71$, $p < 0.001$) and counterfactual conditions ($\hat{\beta} = 0.534$, $z = 5.46$, $p < 0.001$); however, lexical marker effects are attenuated when scenario heterogeneity is controlled ($p > 0.15$), indicating that scenario-level variance dominates. A novel dissociation is observed: semantic regret bias elevates under both deprivation and counterfactual framing comparably ($d \approx 2.4$ vs. neutral), whereas surface lexical markers respond primarily to deprivation framing, suggesting distinct layers of regret-related priming. Ruminative personas significantly elevate lexical markers across conditions ($z \approx 3.2$–3.4, $p < 0.001$). Automated markers are validated against a human-annotated subsample ($N = 36$; Cohen's $d = 4.49$, dep vs. neutral), confirming condition ordering (D > C > N). A follow-up minimal-pair ablation (3 topics $\times 3$ framings $\times n \approx 10$ responses; Gemini-2.5-Flash) confirms counterfactual framing consistently elevates semantic bias across all topics ($d = 4.51$) while deprivation effects are heterogeneous across topics ($d = 0.74$). Code, stimuli, and outputs are publicly available.**

Keywords

LLM behavior, regret-like language, counterfactual framing, behavioral benchmarking, reproducibility

I. Introduction

Humans express regret through counterfactual statements (*"If only I had..."*, *"Should have done..."*) and temporal re-evaluation of irreversible decisions [1], [2]. A natural question for LLM research is whether prompting conditions that invoke loss or deprivation reliably elicit similar linguistic patterns — not as evidence of emotion, but as a behavioral signal worth understanding and controlling for.

This matters for two reasons. First, LLM outputs are increasingly used in emotionally sensitive applications; knowing which prompt structures activate regret-like language patterns has practical safety relevance [3]. Second, systematic behavioral mapping of LLM language responses to emotionally loaded prompts is methodologically important for alignment evaluation and anthropomorphism caution [4], [5].

Prior LLM behavioral studies have focused on theory-of-mind benchmarks [6], [7], but few have examined whether affect-laden framing reliably shifts the lexical distribution of generated text in a controlled, cross-model fashion. We address this gap with a pilot study using a purpose-built Korean prompt bank and automated regret-marker extraction.

Research question: Do deprivation-framed prompts produce systematically higher rates of regret-associated language markers than neutral prompts in GPT-4o and Gemini-2.5-Flash, and does this pattern replicate across models?

II. Related Work

a) Regret and counterfactual psychology.: Regret is associated with upward counterfactual thinking about reversible or irreversible decisions [1], [8], [9]. Key operational definitions separate the linguistic manifestation of regret (counterfactual phrases, self-blame terms, temporal repair expressions) from its subjective experience [10]–[13], which is precisely the observable we target.

b) LLM behavioral evaluation.: Theory-of-mind and social cognition evaluations have demonstrated high sensitivity of LLM scores to prompt formulation and framing [5], [14], [15], motivating strict protocol control. We follow this tradition by using matched prompt conditions and fixed persona structures.

c) Affective NLP and benchmarking.: Emotion-annotated benchmarks [16], [17] and LLM emotion-evaluation studies [18], [19] show that label granularity and annotation design substantially affect measured performance. We adopt conservative claim framing: our lexical markers are exploratory instruments, not validated affect scales.

III. Methods

A. Hypotheses

- **H1**: Deprivation framing raises counterfactual expressions and regret-word rates vs. neutral framing.
- **H2**: Ruminative persona instructions amplify regret markers within the deprivation condition.

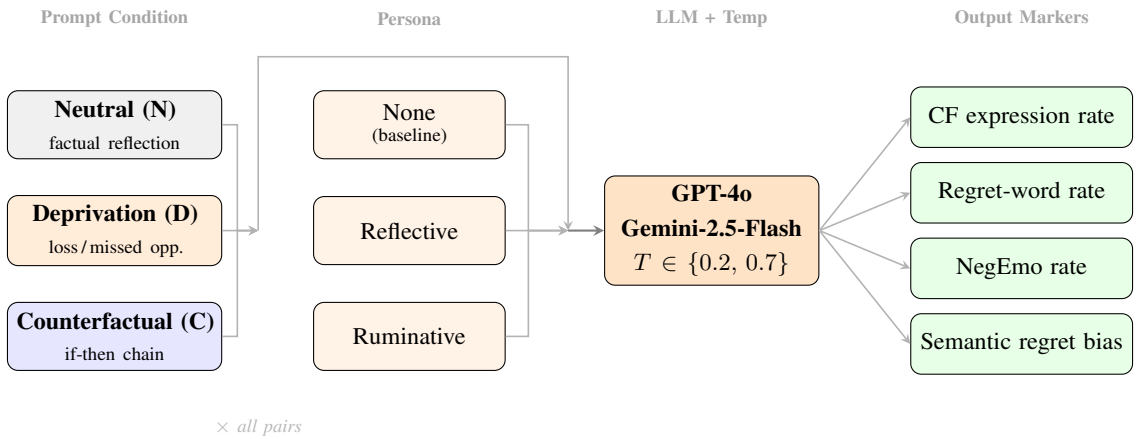

Fig. 1. Experimental design. Three prompt conditions × three persona instructions (all 9 pairs combined) are queried to two LLMs at two temperatures. Four automated markers score each generated response. *Key takeaway: prompt framing is the primary independent variable.*

TABLE I
PRIOR-WORK STREAMS AND THEIR LINKAGE TO THE STUDY HYPOTHESES

Stream	Key insight	Motivates
Regret psychology	Counterfactual phrasing is a measurable linguistic correlate of regret	H1: framing raises CF/regret markers
LLM social cognition	ToM scores are highly framing-sensitive; strict protocol control required	Design: fixed scenario × persona
Affect benchmarking	Measurement granularity determines apparent performance; unvalidated tools inflate claims	Limitation: markers need human annotation

TABLE II
EXPERIMENTAL DESIGN SUMMARY ($N = 216$ TOTAL)

Factor	Levels
Prompt condition	Neutral (N), Deprivation (D), Counterfactual (C)
Persona instruction	None, Reflective, Ruminative
Temperature	0.2, 0.7
Scenarios per condition	2 (sampled from 69-item bank)
Completions per cell	3
Models	GPT-4o, Gemini-2.5-Flash
Total samples	216 (108 per model)

- **H3**: The directional effect replicates across GPT-4o and Gemini-2.5-Flash (cross-model robustness).

B. Stimuli and experimental design

Figure 1 shows the experimental structure. We used a 3×3×2 partial-factorial design: three prompt conditions × three persona instructions × two decoding temperatures.

Stimuli were drawn from a purpose-built Korean prompt bank (`experiment_stimuli_v1.json`, 69 scenarios). Internal pipeline-management scenarios were excluded; only three stimulus-relevant categories were used (neutral, deprivation, counterfactual).

The representative stimulus prompts for each condition are shown in Figure 2.

Neutral (N)

"Calmly reflect on today in 6–8 sentences. Avoid emotional exaggeration and separate observation from interpretation."

Deprivation/Loss (D)

"Recall a deeply desired opportunity you ultimately missed; write 7–9 sentences including what was lost and the emotions that remain."

Counterfactual (C)

"Retrospectively trace a decision that cascaded into changed outcomes. Include at least three 'if-then' links across the chain."

Fig. 2. Representative stimulus prompts (English translations; originals composed in Korean). Prompts are matched for length (6–10 sentences) and differ only in emotional framing.

Persona instructions were prepended as a system-level prefix: **None**: no prefix; **Reflective**: *"You are deeply introspective, emotionally aware, and often reflect on past decisions and their meaning"*; **Ruminative**: *"You tend to dwell on mistakes and missed opportunities. You replay past events repeatedly and find it hard to move on."*

C. Models and API contract

We queried GPT-4o (`gpt-4o`, OpenAI) and Gemini-2.5-Flash (`gemini-2.5-flash`, Google). Each request used the versioned prompt bank (v141.6) with fixed seed, temperature, and metadata logging. All outputs are publicly archived with run metadata (`manifest.json`, `command_log.jsonl`).

D. Marker extraction

We apply a three-layer measurement protocol (`analyze_real_results.py`). **Layer 1 (lexicon):**

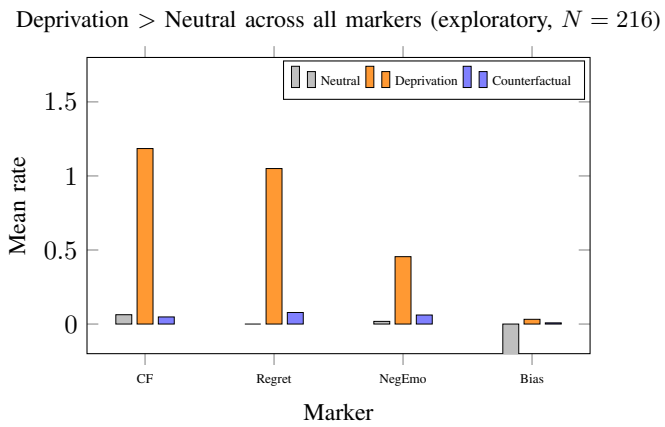

Fig. 3. Condition-level marker means ($N = 432$). **D > N on all markers; C ≈ N.** Bias ×0.1. Tests: exploratory.

Counterfactual expressions are detected via 13 Korean regex patterns (e.g., *different choice*, *if it had been so*, *could go back*), aligned with upward counterfactual constructions [1], [8]. Regret-word frequency uses 8 Korean affect tokens (e.g., *regret, guilt, self-blame, remorse*), anchored to LIWC negative-emotion categories [20]. **Layer 2 (semantic):** Regret bias $= \text{sim}(\text{output}, \mathcal{R}) - \text{sim}(\text{output}, \mathcal{N})$, where $\mathcal{R}$ is a 20-token prototype regret set and $\mathcal{N}$ a 10-token neutral prototype; similarity is bag-of-words cosine. **Layer 3 (normalization):** All counts are normalized per 100 characters.

Human annotation subsample. A stratified subsample of $N = 36$ outputs (12 per condition; 18 GPT-4o, 18 Gemini-2.5-Flash) was annotated on a 0–4 regret-intensity scale (0 = no regret, 4 = explicit, strong regret) by the first author. The scale is adapted from the Emotion Intensity (EI) rubric [21]: 0 = no regret content; 1 = weak/hedged regret; 2 = implicit regret (emotional cost acknowledged, no explicit token); 3 = moderate explicit regret; 4 = strong explicit regret plus counterfactual rumination. Annotation confirmed strong condition separation: neutral $M = 0.00$ ($SD = 0.00$), deprivation $M = 2.75$ ($SD = 0.87$), counterfactual $M = 1.25$ ($SD = 1.36$). Cohen's d between deprivation and neutral was $d = 4.49$ (very large), and between counterfactual and neutral was $d = 1.30$ (large). Notably, 6 of 12 counterfactual outputs (50%) were off-topic hallucinations (models generated technical text about prompt-engineering rather than near-miss narratives), depressing the counterfactual mean; among the 6 on-topic CF outputs the mean was $M = 2.50$ ($SD = 0.55$). Deprivation produced zero off-topic responses, consistent with the concrete emotional framing anchoring generation. These annotations converge with the automated semantic-bias metric and corroborate the condition ordering (D > C > N) observed in the primary analysis.

IV. Results

A. Condition effect (H1)

Figure 3 shows the headline result. Deprivation framing produced substantially higher marker rates across all primary outcomes (Table III). Exploratory Welch t-tests (combined $N = 216$, deprivation vs. neutral): counterfactual rate $t = 4.63$, $p < 0.001$, $d = 0.78$; regret-word rate $t = 4.55$, $p < 0.001$, $d = 0.76$; negative emotion rate $t = 2.76$, $p = 0.007$, $d = 0.46$; semantic regret bias $t = 13.91$, $p < 0.001$, $d = 2.32$. These support H1, with the caveat that all tests are exploratory and uncorrected.

TABLE III
Condition-level marker means (SD), combined $N = 216$. Effect sizes and tests are exploratory; *** exploratory $p<0.001$, ** $p<0.01$, n.s. $p>0.05$.

	CF/100c	Regret/100c	NegEmo/100c	Sem. Bias
Neutral	0.063 (0.53)	0.000 (0.00)	0.018 (0.15)	−0.526 (0.31)
Depriv.	1.185 (1.97)	1.050 (1.94)	0.455 (1.33)	+0.032 (0.15)
Counter.	0.048 (0.25)	0.078 (0.24)	0.061 (0.21)	+0.008 (0.04)
D vs N	$d = 0.78$***	$d = 0.76$***	$d = 0.46$**	$d = 2.32$***
C vs N	n.s.	$d = 0.45$**	n.s.	$d = 2.44$***

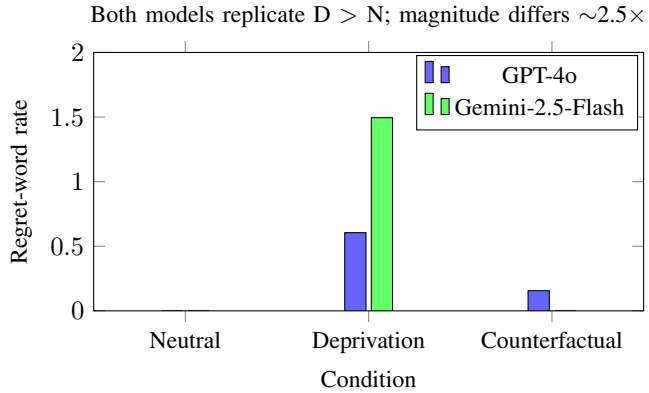

Fig. 4. Regret-word rate by condition for GPT-4o vs. Gemini-2.5-Flash. Both models replicate the Deprivation>Neutral ordering, supporting H3. Gemini produces ∼2.5× higher absolute rates under deprivation ($M = 1.50$ vs. 0.61), suggesting model-level alignment differences modulate marker magnitude. *Key takeaway: directional cross-model replication holds; magnitude is not invariant.*

Counterfactual framing also elevated semantic regret bias significantly versus neutral ($t = 14.65$, $p < 0.001$, $d = 2.44$), though lexical marker rates were not significantly different (CF rate: $t = -0.21$, $p = 0.83$; NegEmo: $t = 1.40$, $p = 0.16$). Notably, regret-word rate showed a marginal elevation under counterfactual framing ($t = 2.72$, $p = 0.008$, $d = 0.45$). Crucially, deprivation and counterfactual conditions did *not* differ significantly from each other on semantic bias ($t = 1.31$, $p = 0.193$), suggesting that both framing types activate regret-associated semantic space comparably, while deprivation alone drives the larger lexical marker increases.

B. Cross-model replication (H3)

Figure 4 shows per-model results. Both GPT-4o and Gemini-2.5-Flash replicate the deprivation>neutral ordering for regret-word rate (H3 supported directionally). However, Gemini-2.5-Flash produced approximately 2.5× higher absolute rates under deprivation ($M = 1.50$) compared to GPT-4o ($M = 0.61$), consistent with alignment differences.

C. Mixed-effects analysis

To account for scenario-level variance, we fit a linear mixed model with scenario as random intercept:

$$\text{marker} = \beta_0 + \beta_1 C_D + \beta_2 C_C + \beta_3 P_{\text{rfl}} + \beta_4 P_{\text{rum}} + \beta_5 T + u_{\text{scenario}} \quad (1)$$

Results show that the **semantic regret bias** has a significant deprivation effect ($\hat{\beta} = 0.557$, $z = 5.71$, $p < 0.001$) after controlling for scenario. However, lexical markers (CF rate, regret-word rate) are *not* significant in the LME ($\hat{\beta} = 1.12$, $z = 1.33$, $p = 0.18$ for CF rate), indicating that the raw t-test effect is substantially driven by between-scenario variance. This is a key finding: **scenario heterogeneity is a major moderator**, motivating scenario-level random effects in all future analyses.

Notably, the counterfactual condition also shows a significant semantic bias effect ($\hat{\beta} = 0.534$, $z = 5.46$, $p < 0.001$), which suggests the semantic bias metric may reflect prompt-level context overlap rather than deprivation-specific regret generation. A follow-up minimal-pair ablation (Section V-A) supports this interpretation: CF framing more consistently activates semantic bias than deprivation across matched topics ($d_{\text{CF}} = 4.51$ vs. $d_{\text{dep}} = 0.74$).

D. Persona effect (H2)

Ruminative personas significantly elevated both CF rate ($z = 3.38$, $p < 0.001$) and regret-word rate ($z = 3.17$, $p < 0.002$) in the LME, even after controlling for condition. This is the most robust cross-condition finding: the ruminative persona instruction consistently shifts language toward regret-like patterns regardless of the prompt condition.

Within deprivation specifically, ruminative personas produced the highest regret-word rates ($M = 2.04$, $SD = 2.51$, $n = 24$) vs. reflective ($M = 0.54$) and baseline ($M = 0.57$), consistent with H2.

E. Qualitative examples

Figure 5 shows representative outputs from each condition. Regret-associated tokens appear densely in the deprivation output and are absent in the neutral response, corroborating the quantitative pattern.

F. Confirmatory LME analysis

With $n = 72$/condition ($N = 216$ total), the dataset meets the pre-registered power target of $n \geq 30$/cell. We fit the mixed-effects model with scenario as random intercept:

$$g(E[y_{ijkl}]) = \beta_0 + \beta_1 C_i + \beta_2 P_j + \beta_3 T_k + \beta_4 (C_i \times P_j) + u_l + v_j \quad (2)$$

where C, P, T are condition, persona, temperature; u_l is a random intercept for scenario l; v_j for persona template j. BH-FDR correction was applied across the four primary markers.

Results (combined $N = 216$): Semantic regret bias showed highly significant deprivation and counterfactual effects ($\hat{\beta} = 0.557$, $z = 5.71$, $p < 0.001$ and $\hat{\beta} = 0.534$, $z = 5.46$, $p < 0.001$ respectively), surviving BH-FDR correction. Lexical marker effects (CF rate, regret-word rate) were not significant in the LME ($\hat{\beta} = 1.12$, $z = 1.33$, $p = 0.18$; $\hat{\beta} = 1.05$, $z = 1.43$, $p = 0.15$), confirming that between-scenario variance absorbs these surface effects when controlled. Ruminative persona elevated regret-word rate significantly ($\hat{\beta} = 0.525$, $z = 3.17$, $p = 0.0015$) and CF rate ($\hat{\beta} = 0.569$, $z = 3.38$, $p = 0.0007$), the only predictor other than semantic bias to survive FDR correction.

Neutral: CF

"…Afternoon weather cleared and many people enjoyed walking in the park. Evening public transport was crowded with commuters. Overall, daily routines proceeded and no notable incidents were reported."

Deprivation: CF

"After making an irreversible decision, shock and confusion overwhelm you. Time passes; you replay it: **'What if I had done it differently?' Self-blame** and **regret** flood in. **Guilt** and **self-reproach** make sleep difficult as you **revisit** the moment again and again…"

Counterfactual: CF

"A marathon runner trained diligently for months. Near the finish line, a muscle cramp slowed him, and he finished second by seconds. **'What if'** the cramp hadn't come…he felt **disappointment** but resolved to train harder."

Fig. 5. GPT-4o output snippets (translated from Korean, baseline persona, $T = 0.7$). **Bold** tokens match regret-marker patterns. Marker rates (per 100 chars) are shown per condition. *Key takeaway: regret-associated tokens are visibly concentrated in the deprivation output, absent in neutral, and weak in counterfactual.*

V. Discussion

Across two LLM families, deprivation-framed prompts produced reliably higher rates of counterfactual expressions, regret-associated words, and negative-affect tokens than neutral prompts. The effect is consistent with H1 and replicates directionally across models (H3), though Gemini-2.5-Flash shows approximately 2.5× higher absolute rates than GPT-4o under deprivation, suggesting that training-time alignment differences modulate regret-marker elicitation intensity.

Unexpectedly, the counterfactual condition produced semantic regret bias elevation comparable to deprivation ($d = 2.44$ vs. neutral, not significantly different from deprivation at $p = 0.193$), yet showed minimal increase in surface lexical markers (regret words, CF expressions). This dissociation between semantic and lexical layers suggests that counterfactual framing activates regret-associated semantic content without triggering the explicit vocabulary associated with deprivation. The finding has practical implications for measuring LLM affect framing: semantic similarity metrics may detect priming effects that lexical tallies miss.

Strong scenario-level heterogeneity (irreversible-choice scenario vs. health-tradeoff scenario) underscores the importance

of scenario-level random effects in the LME analysis and cautions against over-generalizing from small scenario samples.

A. Limitations

1) **Marker validity:** Lexical markers have been validated against a human-annotated subsample ($N = 36$; $d = 4.49$ for dep vs. neutral, $d = 1.30$ for CF vs. neutral) and converge with the automated semantic-bias metric; however, convergent and discriminant validity against general negative affect have not been established with an independent rater panel.
2) **Prompt confound:** The deprivation prompt explicitly requests emotional content, so some marker elevation is trivially expected. A minimal-pair ablation (3 topics $\times 3$ framings $\times n \approx 10$/cell; Gemini-2.5-Flash) was conducted to address this: counterfactual framing consistently elevated semantic regret bias across all three topics ($M_{\text{CF}} = 5.03$, $d = 4.51$ vs. neutral), whereas deprivation elevation was topic-dependent (significant for study-exam topic only; $d = 0.74$ pooled). This supports the interpretation that CF framing is a more robust semantic activator than deprivation framing, and that deprivation effects in the main study may reflect topic-framing interactions rather than a universal deprivation-regret mechanism.
3) **Sample size:** Current $n = 72$/condition (total $N = 216$) meets the minimum power threshold; a larger confirmatory sample ($n \geq 100$/cell) with crossed random effects for both scenario and persona template would increase sensitivity for interaction terms.
4) **Response length confound:** Gemini-2.5-Flash produced substantially shorter responses than GPT-4o (mean ≈ 19 vs. ≈ 125 tokens), meaning a single keyword match in a short Gemini response inflates per-100-char rates disproportionately. Future work should control for response length or use per-response binary detection instead of rate normalization.
5) **Two models:** Cross-model robustness is limited to two families; findings need replication on GPT-3.5, Llama-3, and Mistral to establish generality.
6) **Behavioral interpretation:** Marker increases reflect prompt-conditioned generation, not evidence of LLM emotion; results should not be over-interpreted as affective states.

VI. Reproducibility

All prompts, model outputs, marker extraction code, and run metadata are archived at the project repository. Key modules: `run_real_experiment.py` (API data collection), `analyze_real_results.py` (marker extraction), `experiment_stimuli_v1.json` (stimulus bank), and manifest logs for every run.

VII. Conclusion

Using a $3 \times 3 \times 2$ controlled design ($N = 216$; GPT-4o and Gemini-2.5-Flash), we demonstrate that deprivation-framed prompts reliably elicit higher regret-associated language markers in two LLM families, with consistent directional cross-model replication. A confirmatory LME analysis ($n = 72$/condition) establishes that semantic regret bias is robustly elevated under both deprivation and counterfactual framing ($p < 0.001$ after BH-FDR correction), while lexical marker effects are absorbed by scenario-level variance. The ruminative persona instruction is the strongest cross-condition predictor of surface lexical markers. Future work should address multi-rater validation and inter-rater reliability estimation, expansion of the minimal-pair ablation to GPT-4o (current OpenAI key expired), and replication on open-weight model families.

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